Treatment Guideline for Pediatric Patients with Bacteremia

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Area: Antimicrobial Stewardship
Applicability: UMHS Clinical
Reference: Guideline

Procedure Statement, Purpose, and Scope
This guideline is intended to help guide antimicrobial therapy for patients admitted to pediatric service lines following the results of Gram Stain, Organism Identification (with or without BioFire© molecular resistance results) and Antimicrobial Susceptibilities. Deviation from the recommendations in this guideline may be required for patients with concomitant infections, history of resistant pathogens, or with antimicrobial allergies or intolerance. The recommendations in this guideline reflect susceptibility patterns found at Michigan Medicine and C. S. Mott Children's Hospital.

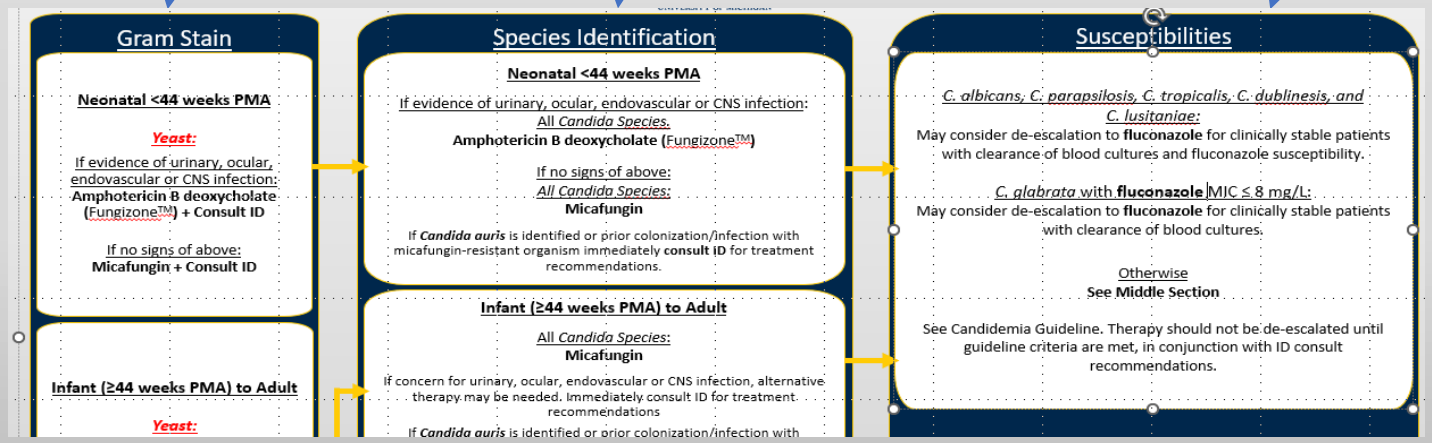
Areas of Responsibility
The guideline will be used by providers and pharmacists caring for pediatric patients with blood stream infections

How to use this guideline:

For patients with ONLY Gram stain results available, refer to the left column (labeled ***Gram Stain***) for treatment recommendations

For patients with organism identification or molecular resistance results available, refer to the middle column (labeled ***Species Identification***) for treatment recommendations

For patients with antimicrobial susceptibility results available, refer to the right column (labeled ***Susceptibilities***) for treatment recommendations



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BCID2 provides identification of the following organisms and is available within ~2 hours from the time of Gram stain on a positive blood culture bottle:

Gram positive[a]	Gram negative[d]	Fungus
Enterococcus faecalis	*Acinetobacter calcoaceticus-baumannii complex*	*Candida albicans*
Enterococcus faecium	*Bacteroides fragilis*	*Candida auris*
Listeria monocytogenes	*Enterobacter cloacae complex*	*Candida glabrata (Nakaseomyces glabrata)*
Staphylococcus aureus	*Escherichia coli*	*Candida krusei (Pichia kudriavzevii)*
Staphylococcus epidermidis	*Haemophilus influenzae*	*Candida parapsilosis*
Staphylococcus lugdunensis	*Klebsiella aerogenes*	*Candida tropicalis*
Staphylococcus spp.[b]	*Klebsiella oxytoca*	*Cryptococcus spp.*
Streptococcus agalactiae	*Klebsiella pneumoniae group*	
Streptococcus pneumoniae	*Neisseria meningitidis*	
Streptococcus pyogenes	*Proteus spp.*	
Streptococcus spp.[c]	*Pseudomonas aeruginosa*	
	Salmonella spp.	
	Serratia marcescens	
	Stenotrophomonas maltophilia	
	Order Enterobacterales[e]	

a) Gram positive organisms not identified by BCID2 include, but are not limited to, *Micrococcus, Peptostreptococcus, Clostridia spp., Bacillus spp. Paenibacillus spp.*

b) Staphylococcus spp. includes, but is not limited to, *S. capitis, S. hominis, S. saprophyticus*

c) Streptococcus spp. includes, but is not limited to, *S. mitis/oralis, S. gordonii, S. bovis*, *S. salivarius, non-Group A/non-Group B β-hemolytic Streptococcus spp.*

d) Gram negative organisms not identified by BCID2 include, but are not limited to, *Haemophilus, Moraxella, Achromobacter*, *non-baumanii Acinetobacter*, *non-aeruginosa Pseudomonas* non-*fragilis Bacteroides spp.* and other Gram-negative anaerobes. **Piperacillin-tazobactam** preferred as empiric treatment

e) Order Enterobacterales includes, but is not limited to, *Citrobacter¸ Morganella, Providencia, Hafnia, Pantoea*

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Gram Stain

Neonatal <44 weeks PMA

Yeast:

If evidence of urinary, ocular, endovascular or CNS infection:
Amphotericin B deoxycholate (Fungizone™**) + Consult ID**

If no signs of above:
Micafungin + Consult ID

Infant (≥44 weeks PMA) to Adult

Yeast:

Micafungin + Consult ID

If suspicion for Cryptococcus or Histoplasmosis (fungemia in setting of pneumonia or meningitis in immuno-compromised patient), call infectious diseases consult service for immediate antifungal recommendations.

Species Identification

Neonatal <44 weeks PMA

If evidence of urinary, ocular, endovascular or CNS infection:
All *Candida Species.*
Amphotericin B deoxycholate (Fungizone™**)**

If no signs of above:
All Candida Species:
Micafungin

If ***Candida auris*** is identified or prior colonization/infection with micafungin-resistant organism immediately **consult ID** for treatment recommendations.

Infant (≥44 weeks PMA) to Adult

All *Candida Species*:
Micafungin

If concern for urinary, ocular, endovascular or CNS infection, alternative therapy may be needed. Immediately consult ID for treatment recommendations

If ***Candida auris*** is identified or prior colonization/infection with micafungin-resistant organism immediately **consult ID** for treatment recommendations.

Cryptococcus spp.
Liposomal amphotericin B (Ambisome™) **+ flucytosine**

Histoplasma:
Liposomal amphotericin B (Ambisome™)

Susceptibilities

C. albicans, C. parapsilosis, C. tropicalis, C. dublinesis, and C. lusitaniae:
May consider de-escalation to **fluconazole** for clinically stable patients with clearance of blood cultures and fluconazole susceptibility.

C. glabrata with **fluconazole** MIC ≤ 8 mg/L:
May consider de-escalation to **fluconazole** for clinically stable patients with clearance of blood cultures.

Otherwise
See Middle Section

See Candidemia Guideline. Therapy should not be de-escalated until guideline criteria are met, in conjunction with ID consult recommendations.

Cryptococcus spp.:
Fluconazole may be appropriate for step down therapy when criteria is met in conjunction with ID consult recommendations

Histoplasma:
Step down therapy may be appropriate when clinical stable in conjunction with ID consult recommendations

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Gram Stain

Gram-positive cocci in clusters*:
Vancomycin

*Single positive cultures from peripheral sources inpatients **without** *Neutropenia, Prosthetic material infection or Premature neonates* may represent contamination; assess for possible source of infection and hold antibiotics if clinically stable

Species Identification

S. aureus or S. lugdunensis and MecA/C + MREJ negative:
Endocarditis or CNS infection: **Nafcillin**
Other Infections: **Cefazolin**

S. aureus or *S. lugdunensis* and *MecA/C* + MREJ positive or *MecA/C* + MREJ not performed:
Vancomycin + Consult ID

Consider discontinuing adjunctive gram-negative therapy between 48-72 hours if cultures are negative for gram-negative pathogens, except for patients with intra-abdominal infection

Single positive culture for Coagulase-negative *Staphylococcus* or *S. epidermidis* in suspected infection of prosthetic material, neutropenia, premature neonates, or in hemodynamically unstable patients:

S. epidermidis and mecA/C Negative:
Cefazolin

S. epidermidis and mecA Positive OR coagulase negative *Staphylococcus*:
Vancomycin

For patients who do not meet the above criteria, a single positive culture for Coagulase-negative *Staphylococcus* or *S. epidermidis* may represent contamination, assess for possible source of infection and hold antibiotics if clinically stable.

Susceptibilities

S. aureus or S. *lugdunensis* sensitive to methicillin:
Endocarditis or CNS infection: **Nafcillin**
Other Infections: **Cefazolin**

S. aureus or S. *lugdunensis* intermediate or resistant to methicillin:
Vancomycin

Coagulase-negative Staph or *Staph epidermidis* sensitive to methicillin:
Endocarditis or CNS infection: **Nafcillin**
Other Infections: **Cefazolin**

Coagulase-negative Staph or *Staph epidermidis* intermediate or resistant to methicillin:
Vancomycin

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Gram Stain

Gram-positive cocci in chains or pairs:

Vancomycin

Species Identification

E. Faecalis and VanA/B Negative:
Ampicillin (Consider **ampicillin/sulbactam** or **piperacillin/tazobactam**)
PCN allergy: **Vancomycin**

E. Faecalis and VanA/B Positive:
Ampicillin (Consider **ampicillin/sulbactam** or **piperacillin/tazobactam**)
PCN allergy: **Linezolid**
[**Daptomycin** for BMT patients pre-engraftment]

E. Faecium and VanA/B Negative:
Vancomycin

E.Faecium and VanA/B Positive:
Linezolid
[**Daptomycin** for BMT patients pre-engraftment]

E. Casseliflavus or E. gallinarium:
Linezolid
[**Daptomycin** for BMT patients pre-engraftment]

S. pneumoniae, S anginosus or Strep. Species:
Most Infections: **Ceftriaxone***
Meningitis/endocarditis: **Vancomycin + Ceftriaxone***
Febrile Neutropenia: **Vancomycin + anti-pseudomonal beta-lactam**

S. Agalactiae or S. pyogenes:
Penicillin or **Ampicillin**
neonates should receive **cefotaxime in place of ceftriaxone*

Susceptibilities

Penicillin-based antibiotics should be first line therapy for all *Enterococcus spp.* if sensitive:
Ampicillin
(Consider **ampicillin/sulbactam** or **piperacillin/tazobactam** for intra-abdominal infections)
Patients w/ PCN allergy or ampicillin-resistant *Enterococcus*:
Vancomycin

Patients with vancomycin allergy (not infusion reaction) or ampicillin and vancomycin-resistant *Enterococcus:*
Linezolid
[Daptomycin for BMT patients pre-engraftment]

Patients with suspected endocarditis will likely require combination therapy and ID consult is strongly recommended

Penicillin-based antibiotics should be first line therapy for all *Streptococcus* spp. if sensitive:
Penicillin or **Ampicillin**
Febrile neutropenic patients should be continued on an **anti-Pseudomonal beta-lactam**

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Gram Stain

Gram-negative rods:
Cefepime (add **metronidazole** for intra-abdominal infections)

*Evaluate if patient has history of resistance to piperacillin-tazobactam or cefepime within prior year, and modify therapy accordingly

.

Species Identification

E. coli, K. pneumoniae, K. oxytoca Proteus, Serratia, Morganella, Enterobacterales*:*

No *CTX-M, KPC, IMP, VIM, NDM, OXA* detected: **Cefepime**
CTX-M positive: **Meropenem**
KPC positive: **Meropenem-Vaborbactam**
IMP, VIM or *NDM (non-E. coli)* positive: **Ceftazidime-avibactam + Aztreonam**
NDM positive *E.-coli*: **Ceftazidime-avibactam + Aztreonam + Tigecycline**
OXA positive: **Ceftazidime-avibactam**

Salmonella spp.

No *CTX-M, KPC, IMP, VIM, NDM, OXA* detected: **Ceftriaxone**
CTX-M positive: **Meropenem**
KPC positive: **Meropenem-Vaborbactam**
IMP, VIM or *NDM* positive: **Ceftazidime-avibactam + Aztreonam**
OXA positive: **Ceftazidime-avibactam**

Enterobacter cloacae complex, Citrobacter freundii, Klebsiella aerogenes

No *CTX-M, KPC, IMP, VIM, NDM, OXA* detected: **Cefepime**
CTX-M positive: **Meropenem**
KPC positive: **Meropenem-Vaborbactam**
IMP, VIM or *NDM* positive: **Ceftazidime-avibactam + Aztreonam** *OR* **Cefiderocol**
OXA positive: **Ceftazidime-avibactam**

Susceptibilities

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.

Narrow-spectrum antibiotics are preferred if no resistance or allergies. These include **ampicillin, penicillin, ampicillin/sulbactam, cefazolin** and **cefuroxime.**

ID consult is strongly encouraged for patients with infections from organisms with CTX-M, *KPC, IMP, VIM, NDM*, or *OXA* resistance genes

Serratia is at low risk for producing an inducible AmpC B-lactamase but caution should still be used when using ceftriaxone or piperacillin/tazobactam as definitive therapy in infections with high organism burden or uncontrolled source (ie: pneumonia, meningitis, or CLABSI w/ retained line)

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.

ID consult is strongly encouraged for patients with infections from organisms with CTX-M *KPC, IMP, VIM, NDM*, or *OXA* resistance genes

Enterobacter cloacae, Citrobacter freundii, and *Klebsiella aerogenes* frequently have an inducible beta-lactamase resistance gene (AmpC), which can confer resistance to penicillin, ampicillin, ampicillin/sulbactam, and 1st-3rd generation cephalosporins. **Cefepime** should be first-line therapy if susceptible.

Citrobacter koseri is not associated with having *AmpC* gene, and narrow spectrum antibiotics should be prescribed if susceptible

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Gram Stain

Gram-negative rods:

Cefepime (add **metronidazole** for intra-abdominal infections)

*Evaluate if patient has history of resistance to piperacillin-tazobactam or cefepime within prior year, and modify therapy accordingly

Species Identification

Pseudomonas aeruginosa

No *IMP, VIM, NDM* detected:
Cefepime

IMP, VIM or *NDM* positive:
Cefiderocol + IV tobramycin until susceptibilities result

CTX-M, KPC, or OXA positive:
Contact infectious diseases – unusual genotype

Acinetobacter baumannii

No *IMP, VIM, NDM* detected:
Cefepime until susceptibilities result

IMP, VIM or *NDM* positive:
Cefiderocol + Minocycline until susceptibilities result

CTX-M, KPC positive:
Sulbactam-durlobactam
Contact infectious diseases – unusual genotype

Susceptibilities

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.

If *Pseudomonas* isolate is resistant to **ceftolozane-tazobactam, ceftazidime-avibactam, or imipenem-cilastatin-relebactam**, request cefiderocol susceptibilities from microbiology lab (phone number 6-6831)

Double coverage of *Pseudomonas* is not indicated after susceptibilities are available, unless isolate is resistant to all beta-lactam antibiotics, cystic fibrosis patient, or decompensating on susceptible antibiotics

ID Consult is encouraged for patients with *Pseudomonas* bacteremia

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.

There is no evidence double coverage of Acinetobacter improves outcomes. The decision to double cover should be made based on source of bacteremia, severity of infection, and patient's medical history

Consider definitive combination therapy for carbapenem-resistant *A. baumannii* based on susceptibility information and infectious diseases consultation

Non-baumannii Acinetobacter spp (e.g., iwofii, radioresistans, etc.) will initially appear as Gram negative rod with a negative Biofire panel and should be treated empirically with **cefepime** or **piperacillin-tazobactam**

Call lab to request susceptibility testing for sulbactam-durlobactam (phone number 6-6831)

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Gram Stain

Gram-negative rods:
Cefepime (add **metronidazole** for intra-abdominal infections)

*Evaluate if patient has history of resistance to piperacillin-tazobactam or cefepime within prior year, and modify therapy accordingly
.

Gram-positive rods:
Most likely the result of skin flora contamination of blood culture

Consider treatment in neonates, HD unstable, prosthetic material with suspected infection, BMT, Neutropenia:
Vancomycin

If concern for Listeria:
Ampicillin
.

Species Identification

Achromobacter:

Piperacillin/tazobactam
PCN allergy: **Meropenem**
(Avoid cefepime unless susceptibility is verified)

Stenotrophomonas maltophilia :

Not neutropenic or critically ill due to S. maltophilia:
Trimethoprim-sulfamethoxazole
Sulfa allergy: **Levofloxacin + minocycline**

Neutropenic or critically ill due to S. maltophilia:
Ceftazidime-avibactam + aztreonam OR **Cefiderocol +/- minocycline**

Bacillus, Lactobacillus and Corynebacterium spp.
Possible contaminants

: Consider treatment in neonates, hemodynamically unstable, prosthetic material with suspected infection, BMT, or Neutropenia

Bacillus or *Corynebacterium spp:* **Vancomycin**

Lactobacillus: **Piperacillin/tazobactam**

Listeria: **Ampicillin**
PCN Allergy: **Meropenem**

Patients with multiple positive sets of blood cultures are more likely true infection. Consider ID consult

Susceptibilities

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.
Achromobacter is frequently multi-drug resistant, and ID consult is encouraged to guide appropriate management of these infections

Piperacillin-tazobactam and cefepime DO NOT have activity against *Stenotrophomonas*

Request cefiderocol susceptibility testing from the clinical microbiology laboratory (6-6831)

Narrow antibiotic selection based on susceptibility results, clinical status, concomitant infections.

Susceptibilities will not be routinely performed by the microbiology lab. Please call to request susceptibilities if strong suspicion for infection (6-6831)

Daptomycin should not be used as definitive therapy for *C. striatum* due to rapid development of resistance. It should be used with caution in all other gram-positive rods.